

Overview

The face occlusion task was treated as a supervised image regression problem. Given a cropped face image, the model predicts a continuous score `FaceOcclusion` $\in [0, 1]$. In order to diversify our explorations and increase the chances of better results, two separate approaches were explored in parallel.

The first approach relies on CNN parameter tuning, with relevant code available at: `datachallenge_resnet50`

The second (and main one with final model) approach was developed with a weighted ensemble of fine-tuned DINOv3 ViT-B/16 models. The final selected submission was `ensemble_final_bestguess_no_run001_run002_mesogip.csv`. It reached a public leaderboard score of **0.00109**. The corresponding code and reproduction instructions are hosted in the project repository: **Final model repository:** `datachallenge_ensemble_dinov3`

1 First approach - CNN parameter tuning

1.1 Model and Architecture

The face occlusion task was treated as a supervised image regression problem using a **ResNet-50** backbone modified for **multi-task learning**. To optimize for both accuracy and demographic fairness, the shared convolutional feature extractor branches into two distinct multi-layer perceptron (MLP) heads:

- **Occlusion Head:** Predicts the continuous `FaceOcclusion` score $\in [0, 1]$.
- **Gender Head:** An auxiliary classification task ($d_{\text{out}} = 1$) outputting raw logits to explicitly model and mitigate sensitive demographic bias.

1.2 Multi-Task Loss Strategy

The joint optimization objective balances accuracy and fairness via a weighted loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{occlusion}} + \gamma_{\text{gender}} \cdot \mathcal{L}_{\text{gender}} \quad (1)$$

Where $\mathcal{L}_{\text{occlusion}}$ is computed via Mean Squared Error (`nn.MSELoss`) and $\mathcal{L}_{\text{gender}}$ utilizes Binary Cross-Entropy with Logits (`nn.BCEWithLogitsLoss`). The hyperparameter γ_{gender} acts as a fairness penalty controlling the strict trade-off between regression performance and demographic regularisation.

1.3 Empirical Tuning and Optimization Loop

Hyperparameters were iteratively tuned on a Slurm cluster. Early baselines with a high penalty ($\gamma_{\text{gender}} = 0.15$) suffered from severe cross-task gradient tension, limiting the local validation score to 0.00135. To resolve this, two optimal configurations were evaluated:

- **First Configuration (50 epochs):** Integrated a robust geometric data augmentation pipeline (`RandomResizedCrop`, `RandomRotation`, `ColorJitter`, `RandomErasing`) as a regularizer. Keeping $\gamma_{\text{gender}} = 0.15$, it reached a local metric of 0.00122 (epoch 39).
- **Second Configuration (150 epochs):** Lowered the penalty to $\gamma_{\text{gender}} = 0.00375$. Relieving the constraint allowed deeper primary task convergence, achieving our best single CNN local score of 0.00112 (epoch 79).

1.4 Leaderboard Results and Post-Hoc Ensembling

On the public leaderboard, a minor distribution shift inverted individual model behaviors: the first configuration scored 0.00125, while the deeper second configuration yielded 0.00129.

Exploiting this variance, a post-hoc arithmetical ensemble (*blending*) was computed by averaging both prediction matrices. This strategy successfully canceled out uncorrelated errors and localized task-biases, achieving a final consolidated public leaderboard score of 0.00117.

2 Second and main approach - Dinov3 ensembling

Model selection

Several backbones were tested sequentially : MobileNetV3-Small/Large, EfficientNet-B0/B1, ResNet-50, ConvNeXt-Tiny and DINOv2-Small/Base. The final model family was DINOv3, specifically `dinov3-vitb16`, which provided strongest pretrained visual representations and best empirical performance. Each backbone was followed by a regression head ending with a sigmoid activation to output in the $[0, 1]$ range. The regression head evolved from a simple linear layer to a deeper MLP head with linear layers, batch normalization, ReLU activations, dropout, and a final sigmoid.

Training strategy

The training strategy was designed to adapt a strong pretrained representation to the face-occlusion task while limiting overfitting. Early experiments kept the full backbone frozen and trained only the regression head. Later experiments used partial fine-tuning by freezing most of the DINOv3 backbone while unfreezing the last transformer blocks. The final ensemble components use a four or six unfrozen blocks unfrozen along with one full fine-tuning run for model diversity. The base loss was a weighted MSE, giving more importance to higher occlusion values to which we added, just as in the first approach, a second weighted loss term for the gender disparity. The best values found empirically for λ_{gap} were around 0.02 to 0.05. The learning rate was scheduled with a cosine decay, and weight decay was used for regularization. The best runs used small learning rates, typically between $2 \cdot 10^{-5}$ and $3 \cdot 10^{-5}$ for partial fine-tuning, and $8 \cdot 10^{-6}$ for full fine-tuning. Finally, we added data augmentation yet to a deliberately moderate extent: horizontal flipping, color jitter, affine transformations, normalization, and random erasing. Test-Time Augmentation (TTA) was also enabled for final predictions, yet with limited performance gains.

Validation and ensembling

The train/validation split was stratified jointly using occlusion level and gender. Multiple random seeds were explored. A key empirical observation was that the best local-validation single model did not necessarily give the best public leaderboard result. For example, several seed-777 DINOv3 models achieved very strong local validation scores, but the public leaderboard improved more when these models were combined with older or more diverse runs. As a result, the final solution was not a single checkpoint but a post-hoc weighted ensemble of six CSV submissions. This ensemble was designed to combine strong seed-777 DINOv3 models with additional diversity from an older DINOv3 run and a full fine-tuning run. The final prediction is computed as: $\hat{y}_{\text{final}} = 0.21\hat{y}_{18142} + 0.18\hat{y}_{18141} + 0.17\hat{y}_{18127} + 0.24\hat{y}_{\text{run002}} + 0.14\hat{y}_{18125} + 0.06\hat{y}_{18138}$, and yields therefore a reproducible CSV-level rather than a PyTorch checkpoint.

Run	GPU	Seed	Epochs	LR	λ_{gap}	Total params	Trainable params	Val. metric	Weight
18142	L40S	777	20	$3 \cdot 10^{-5}$	0.02	85.9M	28.6M	0.001061	0.21
18141	L40S	777	24	$2 \cdot 10^{-5}$	0.05	85.9M	28.6M	0.001063	0.18
18127	L40S	777	20	$3 \cdot 10^{-5}$	0.05	85.9M	28.6M	0.001063	0.17
run002	N/A	N/A	50	$5 \cdot 10^{-6}$	N/A	85.9M	28.6M	0.001616	0.24
18125	H100	123	20	$8 \cdot 10^{-6}$	0.05	85.9M	85.9M	0.001138	0.14
18138	L40S	123	20	$3 \cdot 10^{-5}$	0.05	85.9M	28.6M	0.001157	0.06

Table 1: Final weighted ensemble and its DINOv3 ViT-B/16 predictions constituents

Compute and reproducibility

Experiments were run on several computing environments: a local GPU for early baselines, the Télécom Paris GPU cluster, an external H100 RunPod instance, and the Mesogip/ENSTA cluster using H100 NVL and L40S GPUs for the final runs. Slurm scripts were used for cluster execution. Run logs, hyperparameters, checkpoints, and generated submissions were saved systematically. The naming logic was also adapted to avoid checkpoint collisions when several Slurm jobs were launched in parallel. Overall, in order to obtain the final ensembling model, 6 runs of approximately 2-3 hours are required, therefore a total of 15 hours of compute is approximately required (yet can be parallelized).

Limitations

The final approach remains empirical. No exhaustive Optuna-style hyperparameter search was performed, and the ensemble weights were selected through validation results and public leaderboard feedback. The validation split was useful but imperfectly aligned with the public test distribution, which motivated ensembling but also introduced a risk of overfitting to the public leaderboard. Further improvements would likely come from more advanced ensembling, stronger backbone diversity, more systematic tuning of the gender-gap penalty, pseudo-labeling, or calibration methods designed to handle distribution shift.